Whole Semantic Sparse Coding Network for Remote Sensing Image–Text Retrieval

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***Abstract*—In recent years, cross-modal text-image retrieval in remote sensing (RS) has gained prominence as a research focus due to its potential to provide abundant, inclusive, and multiperspective information. However, existing methods usually focus on salient features, but these salient features cannot describe the image or text completely, resulting in the loss of some important details and discriminable information. In this article, a whole semantic sparse coding network (WSSCN) is proposed for RS image–text retrieval to build a complete and reliable features description for further improving the performance of the retrieval model. Specifically, the WSSCN first designs a whole semantic sparse representation coding (WSSRC) module by constructing a robust semantic library to transform the dense features matrix of the image and text into a whole semantic sparse matrix that enables multiple semantic decoupling and leads to a more precise and detailed features expression. Afterward, the intramodal and intermodal consistency (IIMC) module is devised to improve the intramodal and intermodal consistency of the whole semantic sparse representation from different models. Finally, the salient and whole semantic adaptive learning (SWSAL) loss is proposed to focus on salient information or whole semantic information by calculating an adjusted parameter. Quantitative and qualitative experiments are performed on four extensive datasets for cross-modal retrieval in RS to showcase the notable effectiveness achieved by implementing a WSSCN.**

***Index Terms*—Cross-modal retrieval, remote sensing (RS), sparse coding, whole semantic.**

I. Introduction

THE rapid expansion of satellite sensors has led to a swift increase in both the quantity and quality of remote sensing (RS) data, which finds widespread applications across various fields [1], [2], [3]. To enhance the efficiency of utilizing RS data, numerous studies have focused on retrieval tasks in RS [4], [5], [6]. In contrast to unimodal retrieval in RS, multimodal retrieval holds greater significance as it can extract more abundant semantic information and offer a more comprehensive representation. In particular, the cross-modal retrieval linking images and text is pivotal in multimodal retrieval. This approach possesses the ability to investigate the potential correlation between image and text data, realizing the complementarity of these two modalities. This has increasingly drawn the attention of researchers.The deep learning-based image–text retrieval framework first utilizes a feature extractor to generate high-level feature representation, and then the generated high-level representations are subjected to similarity measurement to obtain the best matching pair. Different from natural images captured by cameras, RS images have their inherent characteristics, which pose great challenges to image–text cross-modal retrieval. For instance, Yuan et al. [7] introduced multiscale feature extraction technology to address the issue of a large scale span of RS targets, and Yuan et al. [8] developed a lightweight network to minimize occupancy and overhead. In spite of the impressive achievements of the methods mentioned above, they have a tendency to extract global information while overlooking local information. This local information is crucial for RS cross-modal retrieval, as the model requires detailed information to effectively identify the best matching pair.

Therefore, some methods are dedicated to devising a fine-grained image–text retrieval method, which is divided into three different levels: patch level, semantic level, and object level. The Transformer-based methods [9], [10], [11] are a typical representative of local modeling since they split the global data into patch-level representations and adapt the attention mechanism to establish interactive connections between local features. However, the patch-based local segmentation pattern may lead to the fragmentation of objects in the RS image. Semantic-level modeling enables the model to prioritize the semantics of interest, thereby enhancing the feature representation capability. Existing semantic modeling techniques mainly involve interaction learning [12], [13], [14], [15], knowledge supervision [16], [17], [18], and multitask learning [19]. Subsequently, benefiting from the development of Fast R-CNN [20], RS images could be represented at a more refined level, specifically at the object-level mentioned above.

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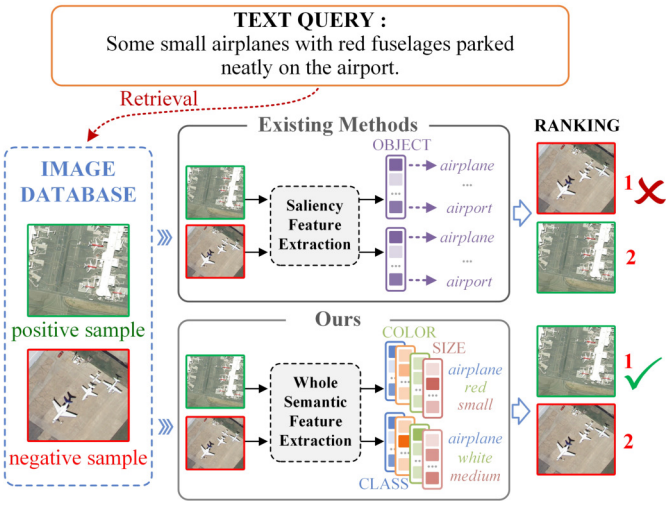


Fig. 1. Comparison of existing methods and ours. The existing methods are dedicated to exploring the salient information of the RS object, but this salient information cannot describe the object completely, eventually leading to an inaccurate ranking outcome. Ours takes into account the whole semantic information, which enables a more comprehensive representation of the RS object, finally outputting the correct ranking result.

For instance, GaLR [21] used the instance features extracted by the object detection network to guide the training of global features; these combination and complementarity boost the model to learn more reliable representations.

Although the previous methods have shown significant progress, they are regarded as an insufficient feature extraction manner, as they only capture significant or partial information about RS objects. However, the RS data hold substantial differences from ordinary data, such as high similarity, small objects, and low resolution. These insufficient feature extraction often ignore some important details and lead to the data losing its discriminability, ultimately limiting the improvement of retrieval accuracy. Here, we show Fig. 1 for an explanation, which illustrates the process of retrieving the image from the database by a given query text. We can observe from the figure that existing methods, like Transformer-based methods [10], [22], apply an attention mechanism to mine salient information in images, but this salient information cannot describe the object completely, eventually leading to an inaccurate ranking outcome.

To address these limitations, we introduce a whole semantic sparse coding network (WSSCN) for enhancing the performance of RS image–text retrieval models. We still adopt the traditional image–text retrieval framework to extract high-level feature representation for images and texts, respectively. In particular, we innovatively design three modules, named whole semantic sparse representation coding (WSSRC) module, intramodal and intermodal consistency (IIMC), and salient and whole semantic adaptive learning (SWSAL) loss, to achieve complete information extraction and matching, so as to ensure effective and reliable image–text retrieval in dealing with complex RS data. Specifically, the WSSRC designs a whole semantic library that contains multiple semantic stacks, such as color, texture, semantics, and relations. Based on this library, the WSSRC applies a kind of cross-attention mechanism and makes use of various semantic information in the library to transform the dense feature matrix of the image and text into a whole semantic sparse matrix that enables multiple semantic decoupling and leads to a more precise and detailed feature expression. The generation and update of this library are realized through a preservation mechanism and distance constraint method to jointly consider both current and historical information and realize the distinction of various semantics in the library. Besides, the IIMC is designed to measure the intramodal and intermodal consistency based on a whole semantic sparse representation. This module first computes intermodal and intramodal saliency matrices, and these saliency matrices are then transferred to bias calculation to generate cross-modal bias; the bias is involved in the similarity calculation process to improve the cross-modal consistency of the matched pairs. Finally, according to the degree of similarity or the category of cross-modal data, SWSAL loss is designed to allow the model to adaptively focus on salient information or whole semantic information by computing an adjustable parameter, achieving a balance between global class and object local details. As shown in Fig. 1, our method pays attention to the whole semantic information, like color, size, and class. This whole semantic information could represent images thoroughly and completely, enabling the model to produce accurate ranking results even with highly similar data. Experiments on four RS image–text retrieval datasets validate the efficacy of the proposed architecture. This article contributes in the following ways.

1) We propose a WSSCN by introducing a whole semantic library to achieve complete information extraction and matching, so as to ensure effective and reliable image–text retrieval in dealing with complex RS data.
2) We propose a WSSRC module by making use of various semantic information in a whole semantic library to enable multiple semantic decoupling and a more precise and detailed feature expression. Besides, we design an IIMC module to measure the intramodal and intermodal consistency from different models. Finally, we present an SWSAL loss to achieve a balance between global semantics and local details.
3) We confirm the effectiveness of our proposed method by evaluating it on four public datasets for RS image–text retrieval, illustrating the superiority of our approach.

II. Related Works

A. Image–Text Retrieval of RS

RS image–text retrieval is defined as the retrieval network's task to locate the corresponding text (or image) from the database based on the measure of similarity when given a query image (or text). Automatic image captioning, the earliest form of cross-modal retrieval method, has evolved over time to become a mainstream approach in the RS field [23], [24], [25], [26]. Considering the limitations of caption-based RS image retrieval, like generating uncontrollable text description, recent efforts have introduced various cross-modal retrieval methods to address these issues. In 2019, the deep bidirectional triplet network (DBTN) introduced a new dataset,

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called TextRS, for the RS image–text retrieval task, marking the beginning of cross-modal retrieval. Different from natural images captured by cameras, RS images have their inherent characteristics, which pose great challenges to image–text cross-modal retrieval. For instance, Yuan et al. [7] introduced multiscale feature extraction technology to address the issue of a large scale span of RS targets, and Yuan et al. [8] developed a lightweight network to minimize occupancy and overhead.

In addition, there has been a growing interest in utilizing prior knowledge as a guide to extract specific and unique information from RS data due to the high similarity between foreground and background. For instance, previous studies, as described in [16] and [18], applied pretrained image feature extractors and text statistics methods to create prior knowledge, respectively, which is then utilized as a supervisory signal to extract effective semantics from hybrid data. In recent years, the emergence of pretrained large models (like CLIP [27], BLIP [28], and MiniGPT-4 [29]) has opened up new possibilities for image–text retrieval. These large models are advantageous because they learn from a vast amount of annotated data for various tasks, resulting in better generalization and representation abilities. In addition, given the unique nature of RS data, RS large models have been developed, such as RemoteCLIP [30] and GeoRSCLIP [31]. As a result, some existing studies [32] have utilized these large models to extract feature representations and devised corresponding mechanisms tailored to specific downstream tasks. Moreover, there are also studies [19], [33] that apply multitask learning modes to categorize various semantic elements in complex RS scenes.

B. Fine-Grained Image–Text Retrieval of RS

In spite of the impressive achievements of the methods mentioned above, they have a tendency to extract global information while overlooking local information. This local information is crucial for RS cross-modal retrieval, as the model requires detailed information to effectively identify the best matching pair. Therefore, existing methods are committed to fine-grained modeling and can be mainly summarized into three categories: patch level, semantic level, and object level.

The Transformer-based methods [9], [10], [11] are typically representatives of the patch-level method since they split the global data into patch-level representations and adapt the attention mechanism to establish interactive connections between local features. However, the patch-based local segmentation pattern may lead to the fragmentation of objects in the RS image. Therefore, some works paid attention to semantic extraction, which is an effective way for enabling models to concentrate on specific pixels of interest by emphasizing different semantics, thereby enhancing the representation capability of features. In RS cross-modal retrieval tasks, one mainstream means for semantic extraction is to derive latent semantics from raw data, including two approaches: intramodal guided extraction and intermodal guided extraction. The former typically involved using an attention mechanism [12] or clustering algorithm [13], [34] to highlight salient semantics of unimodal, while the latter applied a cross-attention mechanism [14], [35] or similarity calculation [15] to discover meaningful information across different modals. The previously mentioned methods primarily depend on calculating feature similarity, which makes it challenging to clearly define semantic boundaries and limits interpretability. To address this issue, supervisory-based methods have been developed. These methods involved creating supervisory signals from raw data or external knowledge and using these signals to guide network mining for meaningful semantics. For instance, in studies [16] and [17], features extracted by neural networks and regions of interest (RoIs) were regarded as supervisory signals, respectively. In study [18], external knowledge was employed to obtain more reliable supervisory signals. In addition, some methods [19] also adopted multitask learning, such as introducing semantic segmentation tasks in cross-modal retrieval tasks to obtain effective semantics in complex data.

Unlike natural data captured by ordinary cameras, RS data, particularly RS images, are extremely complex, involving multiple scenes, large scale spans of targets, multiple target coupling, and exhibiting high interclass similarity and low intraclass similarity. Fortunately, benefiting from the development of Fast RCNN [20], RS images could be represented at the refined object level. As the most representative study, GaLR [21] used the instance features extracted by the object detection network to guide the training of global features; these combination and complementarity boost the model to learn more reliable representations. In addition, RS data often include numerous irrelevant backgrounds and complex targets. Therefore, some researchers have dedicated to understanding the relationships between RS targets by utilizing Transformer, hypergraphs, and other technologies. This includes exploring intramodal correlations [36], [37] as well as cross-modal correlations [10], [17], [38]. By establishing these relationships, it becomes possible to achieve mutual guidance and information interaction between targets, ultimately improving the robustness of feature representation.

Despite the significance and value of the fine-grained image–text cross-modal retrieval methods, they are defined as incomplete feature extraction, such as only mining partial or significant information in RS data. However, the RS data hold complicated characteristics, such as high similarity, small objects, and low resolution, and these incomplete feature extraction methods often ignore some important details, leading to the data losing its discriminability and ultimately limiting the improvement of retrieval accuracy. Therefore, this article aims to address this issue and propose a more effective and reliable image–text retrieval network, namely, WSSCN, by introducing a whole semantic library to achieve complete information extraction and matching.

C. Cross-Modal Alignment in Image–Text Retrieval of RS

Cross-modal data alignment is a classic and challenging problem because of the differences in the inherent nature of data. For example, images and texts have significant variations in format and content. One classic approach [32], [38] involved using interactive guidance of images and texts, along with attention mechanisms and similarity calculation, to achieve data alignment. Besides, certain methods also

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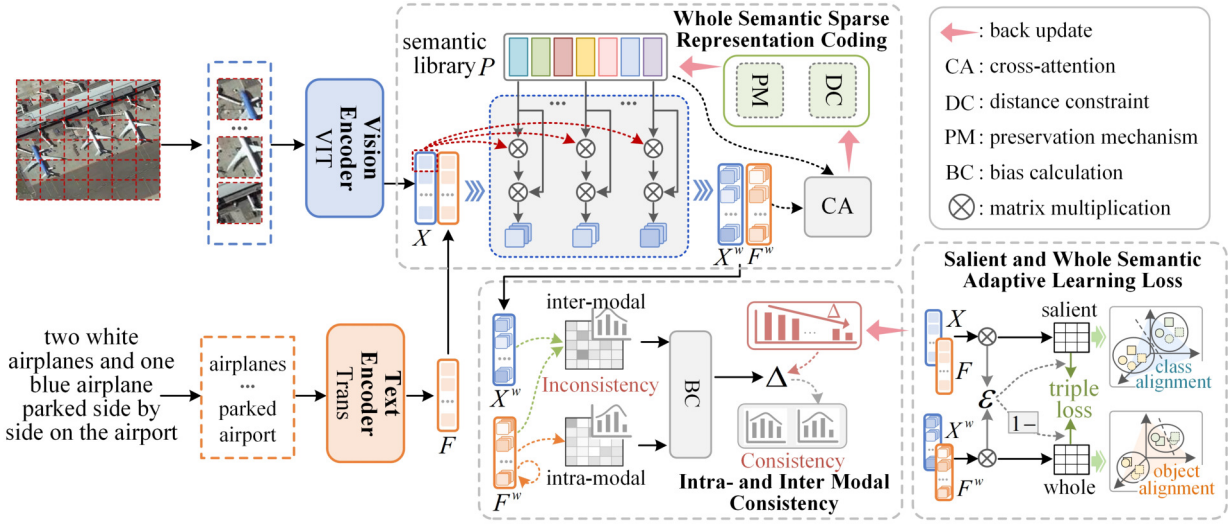


Fig. 2. Architecture of the proposed WSSCN. First, we apply pretrained feature extractors to explore high-level feature representation of the image and text, respectively. The generated features are fed into the WSSRC module, where a whole semantic library is designed with various semantic information to enable multiple semantic decoupling and a more precise and detailed feature expression. Besides, the IIMC is proposed to measure the intramodal and intermodal consistency by reducing the bias of cross-modal saliency matrices during the similarity calculation process. Finally, according to the degree of similarity or the category of cross-modal data, SWSAL loss is designed to allow the model to adaptively focus on salient information or whole semantic information by computing an adjustable parameter.

created intermediate representations that serve as a medium to bring multimodal data closer to it, ultimately reducing the disparity between two distinct modalities. The intermediate representation could be a fusion of multimodal data [14], [39] or a specific region of interest [17]. Although the above exploration can achieve preliminary alignment of modalities, the results are still unsatisfactory in some regions, such as small targets, edges, and fuzzy areas. Hence, some studies [10], [11] tended to utilize Transformer-based techniques, where the patch-level pattern enables the exploration of local detail information while the cross-attention mechanism facilitates the uncovering of potential relationships in the data. The fundamental concept behind these methods is to calculate a similarity matrix and use it to align local representations. To further enhance the interaction between modalities, Zhao et al. [40] designed a masking-based visual-language method, which continuously learns subtle image–text relationships by hiding certain information of the data.

III. Proposed Method

In this section, we introduce the WSSCN, designed to enhance RS image–text retrieval. We begin with an overview of the network, providing insight into its motivation and architecture. Following this, we delve into the detailed description of the proposed modules. Finally, we elucidate the loss function employed for network training.

A. Overview of the Proposed WSSCN

The structure of WSSCN is depicted in Fig. 2. First, we apply pretrained feature extractors to explore high-level feature representation of the image and text, respectively. The generated features are sequentially passed through three modules. The WSSRC module designs a whole semantic library with various semantic information to enable multiple semantic decoupling and a more precise and detailed feature expression. The IIMC is to measure intramodal and intermodal consistency by reducing the bias of cross-modal saliency matrices during the similarity calculation process. Finally, according to the degree of similarity or the category of cross-modal data, SWSAL loss is designed to allow the model to adaptively focus on salient information or whole semantic information by an adjustable parameter.

B. Problem Formulation

As mentioned earlier, the process of cross-modal retrieval is that, given a query image (or text), the retrieval network needs to perform the matching operation with the text (or image) in the database to retrieve the best matching pairs. Here, we formulate the above process for better understanding. Let $\mathcal{D} = \{I, T\}$ be defined as the sample pairs to be matched of images and texts. The WSSCN first applies pretrained feature extractors VIT and Transformer to extract high-level features X and F of the image and text, respectively. Subsequently, the generated image features X and F are fed into the WSSRC module to generate the whole semantic sparse representation X^w and F^w by the guidance of a robust semantic library P. The generation and update of P are realized through a preservation mechanism and distance constraint method to jointly consider both current and historical information and realize the distinction between various semantics in the library. Afterward, WSSCN transmits X^w and F^w into the IIMC module, which initially computes intermodal and intramodal saliency matrices, and these saliency matrices are then transferred to bias calculation to generate cross-modal bias Δ. The bias is involved in the similarity calculation process to improve the intramodal and intermodal consistency of the matched pairs. Finally, according to the degree of similarity or the

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category of X, F and X^w, F^w, the SWSAL loss $\mathcal{L}$ calculate an adjustable parameter ε to allow the model to adaptively focus on salient information or whole semantic information.

C. Feature Extraction

We will detail the feature representation in two aspects, including image feature extraction and text feature extraction.

1) Image Feature Extraction: In recent years, CLIP [27] has exhibited strong feature representation capabilities as a result of its training on extensive datasets, utilizing over 400 million parameters. Furthermore, in response to the unique characteristics of RS data, RemoteCLIP [30] was developed. It is built upon CLIP and trained on 17 diverse and extensive RS datasets across three tasks. Its framework integrates vision Transformer (VIT) [41] for images and Transformer [42] for text, proving to be effective in various downstream RS tasks.

Thus, we utilize pretrained VIT on RemoteCLIP as our image feature extractor. Specifically, the RS image $I \in \mathbb{R}^{H\times W\times 3}$ is first divided into fixed-sized patches. Then these patches are encoded by VIT to get an image feature $X \in \mathbb{R}^{N\times D}$

$$X = \mathcal{F}\left(I, \theta^I\right) \tag{1}$$

where $\mathcal{F}(\cdot)$ denotes the vision encoder and θ^I is the frozen weights.

2) Text Feature Extraction: As previously mentioned, the Transformer trained in RemoteCLIP is very beneficial for text extraction, so we apply it as our text extractor. It is important to note that masked self-attention was utilized in the text encoder to maintain the ability to incorporate language modeling as an auxiliary objective. More specifically, a sentence T is first embedded as word vectors, which is delivered to the Transformer to obtain text feature $F \in \mathbb{R}^{L\times D}$

$$F = \mathcal{T}\left(T, \theta^T\right) \tag{2}$$

where $\mathcal{T}(\cdot)$ denotes the text encoder and θ^T is the frozen weights. Note that Trans in Fig. 2 is the abbreviation of Transformer.

D. Whole Semantic Sparse Representation Coding

The WSSRC is designed to extract a more refined and complete feature representation based on X and F by considering different semantic information of RS. The core idea is to devise a kind of semantic library that contains various semantic stacks, such as shape, color, relation, and texture, and use this library to transform the dense original feature matrix of the image and text into a whole semantic sparse matrix to achieve multiple semantic decoupling, enabling the model to hold the capability of distinguishing complex RS data.

1) Whole Semantic Sparse Representation: Randomly define a whole semantic library P, whose dimension is $P = \{p_1, \ldots, p_c, \ldots, p_C\} \in \mathbb{R}^{C\times M\times D}$, C represents the number of semantics, such as shape, size, and color, and M represents the quantity represented in each semantic, for example, the semantic color is expressed as red, orange, yellow, and green. Subsequently, the library is fully used to transform the original features into whole semantic sparse representations. Here, considering that the cross-attention mechanism in Transformer can aggregate features based on cross-modal correlations, we repeatedly utilize the cross-attention mechanism to increase the feature dimensionality while improving the ability to model relationships between the library P and original features X and F. To simplify the explanation, let us use the image feature X as an example. The implementation process involves considering the image features X as Q in the cross attention and the semantic p_c of the library as K and V

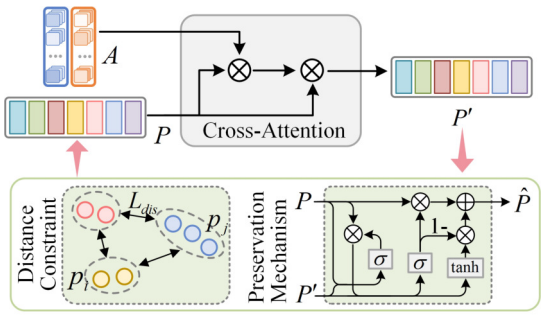


Fig. 3. Generation and update of the whole semantic library, which first utilizes cross attention to update obtained features in the library and then devises a preservation mechanism and distance constraint method to jointly consider both current and historical information and realize the distinction of various semantics in the library.

$$\text{Attention}\,(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

$$x_c^w = \text{Attention}\left(w^{xq}X, w^{pk}p_c, w^{pv}p_c\right) \tag{4}$$

where w_{xq}, w_{pk}, and w_{pv} refer to trainable parameters, $X^w = \{x_1^w, \ldots, x_c^w, \ldots, x_C^w\} \in \mathbb{R}^{C\times N\times D}$. By performing the above operations, the dimension of X changes from 2-D to 3-D, equivalent to decoupling X according to C different semantics. Each $x_c^w \in \mathbb{R}^{N\times D}$ highlights a potentially robust description of semantic c. For example, assuming that the semantic c represents color, then the prominent color is highlighted in N different regions in the feature X^w. Similarly, a whole semantic sparse representation of the text can also be obtained through the above calculation, where $F^w \in \mathbb{R}^{C\times L\times D}$. It can be seen from the above calculation that through sparse modeling, the original features are described in each semantic, achieving a comprehensive and thorough expression of images and texts, thus retaining complete information in complex RS data, including fuzzy and small-scale targets, thereby improving the feature representation capability.

2) Generation and Update of Whole Semantic Library: The whole semantic library has shown powerful strengths that could promote original features to realize semantic decoupling and more fine-grained modeling; thus, it is extremely important to ensure the reliability of library establishment and update. As mentioned above, features X^w and F^w are calculated from original features by highlighting a robust semantic description; thereby, these features could be applied to supplement the library to enrich its semantic representation and improve its reliability and comprehensiveness. In order to update these features in the library, cross attention is again adopted, as shown in Fig. 3. To reduce the computational cost, instead of using each feature to calculate the update

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separately, we first integrate features X^w and F^w by performing a concatenate operation to produce $A = \{a_1, \ldots, a_c, \ldots, a_C\} \in \mathbb{R}^{C\times(N+L)\times D}$

$$A = \left[X^w; F^w\right] \tag{5}$$

where [;] indicates the concatenate operation. Then, we regard p_c of the library as Q in the cross attention and a_c as K and V to transfer the information in A to P and obtain a new library $P' = \{p'_1, \ldots, p'_c, \ldots, p'_C\}$

$$p'_c = \text{Attention}\left(w^{pq}p_c, w^{ak}a_c, w^{av}a_c\right) \tag{6}$$

where w^{pq}, w^{ak}, and w^{av} refer to trainable parameters. Although the library has been updated from the above strategy, there is still a problem that needs to be solved, that is, how to realize the effective combination of the newly generated library P' and the historical library P. It is associated with the preservation mechanism of GRU [43], which selectively retains useful information through the forget gates and hidden gates. Therefore, P continues to be updated through the following formula:

$$\begin{aligned}
\hat{P} &= ZP + (1 - Z)\, H \\
H &= \tanh\left(P' w^{ph} + (RP)\, w^{hh}\right) \\
Z &= \sigma\left(P' w^{pz} + P w^{hz}\right) \\
R &= \sigma\left(P' w^{pr} + P w^{hr}\right)
\end{aligned} \tag{7}$$

where $w^{ph}, w^{hh}, w^{pz}, w^{hz}, w^{pr}$, and w^{hr} refer to trainable parameters and tanh and σ are activation functions. Similar to the gate unit in GRU, H represents the candidate hidden state, R is the reset gate, and Z is the update gate.

In addition, in the whole semantic library, we believe that different semantics have to be distinguishable, for example, the distribution of texture and color must be different. Thus, we adopt mean squared error (mse) loss to ensure the distance constraint between different semantics. Since the mse loss was originally designed to minimize the distance between features, but our goal is to maximize the distance between different semantics, it is formulated in the form of an inverse

$$L_{\text{dis}} = \frac{C^2}{\sum_{j=1}^{C}\sum_{i=1}^{C}\left(p_i - p_j\right)^2}. \tag{8}$$

E. Intramodal and Intermodal Consistency

Although the WSSRC module obtains robust feature descriptions of images and texts, the "cross-modal inconsistency" is still a thorny problem. Taking a positive sample pair as an example, given the text "Four planes are parked at the airport, they are neatly arranged," the "plane" is considered to be the most important word. However, when these words are matched with the image region for cross-modal similarity, the word "plane" may not be as significant as other words, such as "airport." This is the mentioned cross-modal inconsistency, which will cause the original text to lose its important information or discriminability, thereby reducing the matching degree of positive sample pair and outputting incorrect retrieval results. Therefore, in this case, we need to weaken cross-modal inconsistency. The IIMC is designed by calculating a cross-modal bias, which will be reduced in the subsequent similarity calculation and matching process, as depicted in Fig. 4.

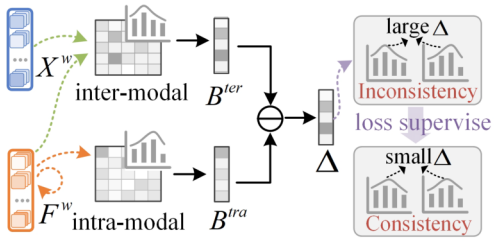


Fig. 4. Framework of IIMC, which first computes intermodal and intramodal saliency matrices, and these saliency matrices are then summed by row and subtracted to generate cross-modal bias, the bias is involved in the similarity calculation process to improve the cross-modal consistency of the matched pairs.

1) Intermodal and Intramodal Saliency Matrices: Based on the obtained whole semantic sparse representation X^w and F^w, we apply saliency matrices to calculate the intermodal and intramodal similarity. It should be noted that RS images contain complex scenes and targets, leading to diverse intramodal saliency areas, while text descriptions are relatively simple and result in a single intermodal saliency. This difference makes it difficult to align the intramodal and cross-modal saliency of the image. As a result, we can only achieve consistency within the text modality and across modalities

$$\begin{aligned}
B^{\text{ter}} &= F^w \times (X^w)^T \\
B^{\text{tra}} &= F^w \times (F^w)^T
\end{aligned} \tag{9}$$

where the B^{ter} and B^{tra} refer the intermodal and intramodal saliency matrices, respectively. Notably, while (9) takes the form of an inner product operation, the resulting matrix functions as a saliency map that quantifies the degree of correlation and alignment between the elements in the two features. Afterward, we perform a summation of each row on matrix B^{ter} and B^{tra}

$$\begin{aligned}
B_i^{\text{ter}} &= \sum_{j=1}^{N} B_{ij}^{\text{ter}} \\
B_i^{\text{tra}} &= \sum_{j=1}^{L} B_{ij}^{\text{tra}}
\end{aligned} \tag{10}$$

where i and j stand for row and column, respectively, and B_i^{ter} and B_i^{tra} are the intermodal and intramodal saliency matrices. Each i in B_i^{ter} refers to the intercorrelation of one word with all other image regions, and each i in B_i^{tra} represents the intracorrelation between one word and other words.

2) Bias Calculation and Updating: After that, we calculate the subtraction between B_i^{ter} and B_i^{tra}, and gain the cross-modal bias Δ

$$\Delta = \sum_{i=1}^{L}\left(B_i^{\text{ter}} - B_i^{\text{tra}}\right). \tag{11}$$

Next, the Δ is involved in the similarity matching process

$$S'\left(X^w, F^w\right) = (1 - \beta|\Delta|)\, S\left(X^w, F^w\right) \tag{12}$$

where $S(X^w, F^w)$ represents the cosine similarity between the whole semantic sparse image and the text features, and

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$S'(X^w, F^w)$ represents the updated similarity score. β is a hyperparameter, controlling how strongly the inconsistency Δ penalizes the original similarity. In the subsequent loss function, $S'(X^w, F^w)$ will be trained with a high score, which means that $(1-\beta|\Delta|)$ will also become higher, so that the value of Δ will decrease accordingly, thereby reducing cross-modal inconsistency.

F. SWSAL Loss

As usual, the triplet loss is applied as a loss function for our cross-modal retrieval model since it can achieve a more refined measure of difference. Triplet loss aims to minimize the distance between the anchor and the positive sample, while maximizing the distance between the anchor and the negative samples, so as to train the network to obtain robust feature representation. Faghri et al. [44] introduced a bidirectional triple loss specifically designed for cross-modal retrieval

$$\begin{aligned}\mathcal{L}(I,T) = &\sum_{\hat{T}} \left[\alpha - S(I,T) + S\left(I,\hat{T}\right)\right]_+ \\ &+ \sum_{\hat{I}} \left[\alpha - S(I,T) + S\left(\hat{I},T\right)\right]_+ \end{aligned} \tag{13}$$

where α refers to the margin and $[\cdot]_+ = \max(\cdot, 0)$. $S(I,T)$ represents the similarity between positive sample pairs of image and text. $S(I,\hat{T})$ is processed for all negative sentences $\hat{F}$ of a given image anchor I and $S(\hat{I},T)$ considers all negative images $\hat{I}$ of a given sentence anchor T.

Here, in order to balance and integrate the salient representation and the generated whole semantic sparse feature, we improve the triplet loss and propose $\mathcal{L}_{sa}$. $\mathcal{L}_{sa}$ is innovative in calculating both positive sample pair similarity and negative sample pair similarity. Specifically, the positive sample pair similarity $S(X,F)$ consists of two parts

$$S(I,T) = S'(X^w, F^w) + S(X,F) \tag{14}$$

where $S'(X^w, F^w)$ is the similarity between the whole semantic sparse features of the image and the text features by integrating the cross-modal bias. $S(X,F)$ refers to the similarity between the original image and the text features. For $S(X,\hat{F})$ and $S(\hat{X},F)$, however, the negative sample pairs include not only different semantics but also the same semantics. As the negative samples sharing the same semantics exhibit high similarity, we believe that the network should prioritize the extraction of detailed local information to uncover subtle distinctions. Conversely, negative samples with different semantics show low similarity, requiring the network to focus on global information for accurate prediction. Therefore, we think of calculating the negative sample pair similarity under semantic awareness. Thus, a parameter ε is computed to learn the proportion of the whole semantic sparse feature similarities and salient feature in the training process so that the model pays attention to the discriminability of the same semantic data. To simplify, we take the calculation of $S(I,\hat{T})$ as an example

$$\varepsilon^x = \frac{S\left(X,\hat{F}\right)}{S'\left(X^w,\hat{F}^w\right) + S\left(X,\hat{F}\right)}. \tag{15}$$

Then, $S(I,\hat{T})$ can be written as

$$S\left(I,\hat{T}\right) = \varepsilon^x S'\left(X^w,\hat{F}^w\right) + (1-\varepsilon^x) S\left(X,\hat{F}\right). \tag{16}$$

Equations (15) and (16) can be interpreted as when X and $\hat{F}$ do not belong to the same semantics, $S'(X^w,\hat{F}^w) > S(X,\hat{F})$, ε^x is smaller at this time, thus modal will focus on the global similarity. When X and $\hat{F}$ have the same semantics, $S'(X^w,\hat{F}^w) < S(X,\hat{F})$, ε^x becomes larger and the role of local similarity also increases. Likewise, for the calculation of $S(\hat{I},T)$, the parameter ε^f is obtained by following the above process. Then, (13) can be written as

$$\begin{aligned}\mathcal{L}_{sa} &= \mathcal{L}(I,T) \\ &= \sum_{\hat{T}} \left[\alpha - \left(S'(X^w,F^w) + S(X,F)\right)\right. \\ &\qquad \left. + \varepsilon^x S'\left(X^w,\hat{F}^w\right) + (1-\varepsilon^x) S\left(X,\hat{F}\right)\right]_+ \\ &\quad + \sum_{\hat{I}} \left[\alpha - \left(S'(X^w,F^w) + S(X,F)\right)\right. \\ &\qquad \left. + \varepsilon^f S'\left(X^w,\hat{F}^w\right) + \left(1-\varepsilon^f\right) S\left(\hat{X},F\right)\right]_+ . \end{aligned} \tag{17}$$

Finally, the loss function is

$$\mathcal{L} = \mathcal{L}_{\text{dis}} + \mathcal{L}_{sa}. \tag{18}$$

IV. Experimental Results and Analysis

In this section, we carried out some experiments to assess the effectiveness of our model. We first describe the datasets, evaluation metrics, baseline, and implementation details. Afterward, we verify the validity of our model through both quantitative and qualitative experiments.

A. Datasets

We perform experiments on four RS datasets for the cross-modal image–text retrieval: UCM-Captions, Sydney-Captions, RSICD, and RSITMD. The UCM-Captions dataset, developed by Qu et al. [45], is an extension of the UCMerced-LandUse dataset [46]. It consists of 2100 images distributed across 21 classes, with spatial dimensions of 256 × 256 pixels. In the study by Qu et al. [45], each image is described using five distinct sentences. The RSICD dataset is built for the RS image captioning task by Lu et al. [47]. It is currently the largest retrieval dataset, including 10 291 RS images containing 33 categories, and each image is described in five sentences. The RSITMD dataset, provided by Yuan et al. [48], is a collection of images specifically selected from the RSICD dataset for cross-modal image–text retrieval research. It contains a total of 23 715 captions corresponding to 4743 images. The captions in the RSITMD dataset are more granular and diverse compared with the RSICD dataset, offering a broader range of descriptive text. Qu et al. [45] have provided the Sydney-Captions dataset [49], which is based on the Sydney dataset. Compared with other datasets, it has a relatively small number of categories, with a total of 613 images divided into seven categories. Each image is accompanied by five sentences for description.

B. Evaluation Metrics

We evaluate our model's performance on two types of image–text matching tasks.

1) *Image-to-Text:* Retrieving relevant sentences for a given query image.

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TABLE I
COMPARISONS OF IMAGE–TEXT RETRIEVAL EXPERIMENTS ON RSICD AND RSITMD DATASETS

Type	Method	Backbone Vision/Text	RSICD			RSITMD		
			Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR	Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR
N	SCAN t2i	ResNet/GRU	4.42 / 11.20 / 17.68	4.02 / 11.54 / 18.60	11.24	10.59 / 28.72 / 38.41	10.04 / 29.54 / 42.91	26.70
	SCAN i2t	ResNet/GRU	5.90 / 13.21 / 19.96	3.86 / 16.83 / 26.49	14.38	10.84 / 25.86 / 37.26	9.62 / 29.80 / 41.06	25.74
	CAMP-triplet	ResNet/GRU	5.22 / 13.05 / 21.02	4.30 / 17.10 / 27.57	14.71	11.82 / 27.40 / 38.12	8.69 / 27.17 / 43.60	26.13
	CAMP-bce	ResNet/GRU	4.50 / 10.08 / 16.48	3.03 / 15.12 / 23.05	12.04	9.21 / 22.56 / 35.73	6.81 / 25.65 / 40.05	23.34
	MTFN	ResNet/GRU	5.01 / 12.86 / 21.30	4.96 / 12.14 / 29.12	14.23	10.80 / 27.68 / 36.40	9.82 / 30.28 / 48.27	27.21
	SGRAF	ResNet/GRU	3.93 / 13.36 / 22.78	3.42 / 14.53 / 26.72	14.12	5.97 / 18.81 / 30.09	6.02 / 24.25 / 41.81	21.16
	NAAF	ResNet/GRU	5.92 / 16.31 / 27.23	4.21 / 17.20 / 26.33	16.20	10.50 / 30.75 / 41.86	9.13 / 30.38 / 44.25	27.81
R	SAM i2t	CNN/RNN	6.41 / 20.10 / 32.89	5.91 / 18.08 / 29.73	18.85	13.27 / 30.09 / 44.91	9.87 / 33.89 / 50.66	30.45
	SAM t2i	CNN/RNN	5.91 / 18.59 / 30.63	4.82 / 18.10 / 33.44	18.58	11.73 / 27.25 / 40.06	11.07 / 35.03 / 50.94	29.35
	LW-MCR	SqueezeNet/RNN	3.66 / 13.63 / 21.68	3.77 / 15.32 / 28.20	14.38	6.64 / 17.92 / 30.31	5.53 / 25.51 / 44.34	21.71
	AMFMN	ResNet/GRU	5.39 / 15.32 / 28.82	5.05 / 18.19 / 29.07	16.97	11.37 / 27.7 / 39.29	9.04 / 32.46 / 49.68	28.26
	GaLR	ResNet+ppyolo/GRU	4.67 / 17.47 / 27.63	4.65 / 20.20 / 34.80	18.24	13.27 / 31.85 / 41.15	10.92 / 35.3 / 52.96	30.91
	MCRN	CNN/GRU	5.58 / 18.38 / 30.56	4.87 / 18.92 / 33.25	18.59	8.85 / 23.67 / 35.84	8.05 / 30.97 / 49.60	26.16
	CMFM	ResNet/GRU	5.40 / 16.47 / 27.54	4.15 / 17.99 / 31.25	17.13	8.85 / 23.89 / 37.17	7.12 / 31.95 / 50.88	26.64
	PIR	Swin Trans/Bert	10.52 / 26.44 / 39.52	6.83 / 23.97 / 39.09	24.40	19.03 / 41.15 / 53.32	13.72 / 42.48 / 62.88	38.76
	MGRM-EL*	ResNet/Skipthought	7.41 / 23.24 / 35.32	5.75/ 21.23 / 35.55	21.42	13.51 / 31.87 / 46.27	11.11 / 37.22 / 56.61	32.76
	DOVE*	ResNet/GRU	8.66 / 22.35 / 34.95	6.04 / 23.95 / 40.35	22.72	16.81 / 36.80 / 49.93	12.20 / 44.13 / 66.50	37.73
	HVSA	CNN/RNN	8.60 / 22.78 / 33.21	5.95 / 22.53 / 34.57	21.27	14.60 / 32.96 / 46.46	10.84 / 38.01 / 57.35	33.37
	KAMCL	ResNet/GRU	8.14 / 22.14 / 33.39	6.35 / 23.70 / 38.87	22.10	14.82 / 29.65 / 43.58	10.53 / 35.8 / 55.84	31.70
	GLISA*	Trans/Trans	19.52 / **40.44** / 52.28	14.75 / 39.50 / 55.46	36.99	28.21 / 50.94 / 62.50	20.50 / 54.80 / 72.19	48.41
	RemoteClip	Trans/Trans	16.65 / 35.13 / 47.39	14.73 / 40.14 / **56.96**	35.17	27.21 / 50.22 / 63.93	24.20 / 57.21 / 74.96	49.62
	IEFT	Trans/Trans	4.86 / 19.76 / 34.48	6.77 / 23.88 / 38.15	21.32	14.82 / 33.63 / 49.56	10.98 / 37.61 / 58.25	34.14
	CLGSA*	Trans/Trans	14.38 / 36.73 / 45.35	11.75 / 34.07 / 50.12	32.07	26.77 / 48.23 / 60.39	22.52 / 52.39 / 69.46	46.63
	AIR*	Trans/Trans	18.85 / 39.07 / 51.78	14.24 / 39.03 / 54.49	36.24	29.20 / 49.78 / 65.27	**26.06** / 57.04 / 73.98	50.22
	WSSCN	Trans/Trans	**19.58** / 39.16 / **51.97**	**14.77** / **40.86** / 56.65	**37.17**	**28.54** / **55.31** / **66.59**	24.29 / **57.43** / **75.35**	**51.25**

2) *Text-to-Image:* Retrieving relevant images for a given query text.

The Recall@K metric (R@K) is adopted to evaluate our model's performance, where K = 1, 5, and 10. R@K measures the proportion of ground-truth results that are present in the top K results returned by the model. In addition, we calculate the mean Recall@K (mR) for R@K, as proposed by Huang et al. [50], to provide a more comprehensive assessment of the model's performance. The mR is calculated by taking the mean of the recall rates for each query and then averaging them. This metric provides a better understanding of the model's performance by considering the average recall for all queries, rather than just the maximum recall for a single query.

C. Implementation Details

In this section, we mainly introduce the implementation details of the proposed method. The dimension D of features is 512, and the number of semantic categories C is 8. The margin α of the triplet loss was set to 0.2. All the models are implemented with PyTorch and the Adam optimizer with a 0.0002 learning rate. We set the batch size of training to 30 and train the model for 50 epochs. All the experiments are conducted on a Tesla V100 SXM2 16-GB GPU.

D. Baselines

We selected a total of 21 image–text retrieval methods for comparison, including five classic image–text retrieval networks, abbreviated as "N" in subsequent experiments, such as SCAN [38], CAMP [51], MTFN [52], SGRAF [53], NAAF [54], and sixteen networks in the RS field, recorded as "*R*," which are SAM [14], LW-MCR [8], AMFMN [7], GaLR [21], MCRN [55], CMFM [56], PIR [16], MGRM-EL [36], DOVE [17], HVSA [57], KAMCL [18], GLISA [58], RemoteCLIP [30], IEFT [10], CLGSA [59], and AIR [60]. It should be noted that since some methods do not have publicly available codes, we directly quote the results provided by the authors in the published manuscript and mark them with the symbol "*."

E. Comparison With State-of-the-Art Methods

In this section, we present a comparative analysis of our approach with 21 baseline models, evaluated on four diverse datasets. The experimental results are summarized in Tables I and II. Given the variance in performance across different datasets, we offer the following in-depth examination.

Results on RSICD: The results on the RSICD dataset are shown in the left of Table I. As can be observed from the table, among all the classic cross-modal retrieval methods, the NAAF achieves the best performance, with an accuracy of 16.20 on the mR evaluation metric. For models in the field of RS, GLISA exhibits the highest cross-modal retrieval result of 36.99 on the mR evaluation metric. The most significant result in the first table is the proposed WSSCN, which achieves a retrieval accuracy of 37.17 of mR, with an improvement of 0.49% compared with the current best method GLISA, thus proving the effectiveness of the proposed model. However, in the text-to-image retrieval task, WSSCN's performance is inferior to RemoteClip's. This is due to RemoteClip's superior

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TABLE II
COMPARISONS OF IMAGE–TEXT RETRIEVAL EXPERIMENTS ON UCM AND SYDNEY DATASETS

Type	Method	Backbone Vision/Text	UCM			SYDNEY		
			Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR	Image-to-Text R@1 / R@5 / R@10	Text-to-Image R@1 / R@5 / R@10	mR
N	SCAN t2i	ResNet/GRU	13.89 / 45.81 / 68.59	13.11 / 50.69 / 78.21	45.05	19.14 / 50.71 / 74.12	17.81 / 55.94 / 76.57	49.05
	SCAN i2t	ResNet/GRU	12.81 / 46.99 / 69.11	12.59 / 46.89 / 72.71	43.52	20.42 / 54.16 / 67.62	16.13 / 57.61 / 75.98	48.65
	CAMP-triplet	ResNet/GRU	11.20 / 44.29 / 65.72	9.89 / 46.11 / 77.29	42.42	22.83 / 50.46 / 75.88	15.31 / 43.14 / 70.39	46.34
	CAMP-bce	ResNet/GRU	15.09 / 47.18 / 68.59	11.93 / 47.22 / 76.98	44.50	15.64 / 49.65 / 71.33	11.59 / 51.34 / 76.16	45.95
	MTFN	ResNet/GRU	10.68 / 47.62 / 64.31	14.74 / 52.65 / 81.13	45.19	21.71 / 51.58 / 69.03	14.13 / 55.96 / 78.62	48.51
	SGRAF	ResNet/Bi-GRU	4.76 / 36.19 / 63.81	10.76 / 48.67 / 78.38	40.43	15.52 / 43.10 / 62.07	7.93 / 46.55 / 73.10	41.38
	NAAF	ResNet/GRU	12.38 / 49.52 / 64.76	11.52 / 44.38 / 66.29	41.48	24.14 / **56.90** / 68.97	17.24 / 46.90 / 75.86	48.34
R	SAM i2t	CNN/RNN	13.78 / 46.77 / 67.15	11.32 / 49.27 / 81.81	45.02	24.12 / 55.17 / **75.93**	13.41 / 54.46 / 76.62	49.95
	SAM t2i	CNN/RNN	11.92 / 45.23 / 70.95	11.42 / 53.41 / 83.12	46.01	**27.64** / 56.87 / 72.35	19.01 / 57.93 / 79.96	52.29
	LW-MCR	SqueezeNet/RNN	13.81 / 40.10 / 59.05	10.38 / 44.86 / 74.95	40.53	17.24 / 43.10 / 63.79	16.21 / 56.21 / 75.17	45.29
	AMFMN	ResNet/GRU	14.48 / 51.17 / 67.35	14.49 / 51.82 / 80.71	46.67	20.69 / 50.73 / 75.23	13.45 / 60.00 / 82.70	50.47
	MCRN	CNN/GRU	12.38 / 44.29 / 68.09	11.71 / 50.48 / 92.76	46.62	25.86 / 55.17 / 65.52	15.52 / 54.83 / 80.34	49.54
	CMFM	ResNet/GRU	13.33 / 50.00 / 66.67	11.43 / 49.24 / 74.57	44.21	17.24 / 51.72 / 62.07	15.17 / 55.52 / 81.38	47.18
	HVSA	CNN/RNN	14.29 / 39.52 / 59.52	13.71 / 48.48 / 73.33	41.48	13.79 / 48.28 / 62.07	14.48 / 55.52 / 77.24	45.23
	RemoteClip	Trans/Trans	15.71 / 50.95 / 80.47	17.71 / 63.33 / **98.00**	54.36	20.69 / 56.89 / 74.14	21.03 / 64.83 / 82.41	53.33
	IEFT	Trans/Trans	11.90 / 47.62 / 75.71	13.63 / 53.03 / 90.45	48.72	22.41 / 63.79 / 72.41	18.59 / 54.65 / 79.55	51.90
	WSSCN	Trans/Trans	**24.76** / **60.48** / **84.76**	**21.43** / **66.57** / 96.19	**59.03**	18.97 / **56.90** / 74.15	**23.70** / **65.86** / **87.24**	**54.48**

generalization ability, achieved through training on large-scale datasets.

Results on RSITMD: From the right of Table I, NAAF obtains the best results among all the classic retrieval models. In the RS field, AIR delivers satisfied output, outperforming all models and achieving an mR accuracy of 50.22. It is evident that the proposed WSSCN demonstrates the best performance on the metric mR. Our approach achieves 84.29% and 2.05% gains compared with the cross-modal retrieval baseline NAAF and AIR in classic and RS, respectively. We attribute these remarkable results to the reliability and stability of the proposed WSSRC, IIMC, and SWSAL loss, providing a complete and comprehensive feature description.

Results on UCM: The experiment results obtained from the UCM dataset are displayed in the right section of Table II. It is noticeable that all models exhibit higher retrieval accuracy compared with other datasets. Among all state-of-the-art methods, RemoteClip obtains the best retrieval precision of 54.36 on the mR evaluation metric due to its robust feature representation and generalization capabilities trained from various and multiple task datasets. Despite the superior performance of RemoteClip, our method outperforms it by 8.59% on the mR evaluation metric because we not only design a whole semantic sparse representation, which is more robust than the mentioned incomplete feature, but also consider the intermodal and intramodal consistency to further explore more effective cues. Moreover, in the metric of R@1, our method far exceeds all baselines, improving by 57.61% and 21.01% compared with RemoteClip in image-to-text and text-to-image tasks, respectively. However, the performance of the R@10 metric in the text-to-image retrieval task is not as good as that of RemoteClip, and we deduce, that is, because RemoteClip has a higher model generalization capability.

Results on SYDNEY: It can be observed in the right of Table II that the SCAN t2i and RemoteClip achieve the best results compared with the classic and RS retrieval networks, yielding 49.05 and 53.33 results on the mR evaluation metrics, respectively. For text-to-image retrieval, the proposed model shows competitive advantages in R@5 and R@10, but is significantly inferior in R@1. For image-to-text retrieval, WSSCN surpasses all the baselines in all evaluation metrics. Nonetheless, the overall performance of WSSCN is still prominent, with a score of 54.48 on the mR evaluation metric. The results on SYDNEY indicate the availability and feasibility of the proposed modules.

F. Ablation Studies

In this section, we perform ablation experiments to explore the effectiveness of the modules presented here. The experimental results are shown in Table III, and in the table, FE means only adopted the feature extraction module, and WSSRC, IIMC, and SWSAL loss represent the proposed three modules. "+" means that the model integrates and adopts the corresponding modules. As we can see, the modal only performed feature extraction exhibits the worst performance, with only 49.62 on the mR evaluation metric. The integration of WSSRC and IIMC modules promotes the performance of the retrieval model, improving by 1.87% and 0.93% compared with FE, respectively, and the WSSRC module has the most significant advantage since it builds a reliable library to transfer incomplete dense feature representation to a whole semantic sparse representation that greatly enhances feature representation capabilities. As the modules continue to be integrated, the accuracy of retrieval is gradually improved. When the model applies all three modules, which is shown in the last row of the table, the model shows the best performance, improving FE by 3.28%. Therefore, the above data demonstrate the effectiveness of the proposed method from a quantitative perspective.

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TABLE III
ABLATION STUDIES

Method	RSITMD						
	Image-to-Text			Text-to-Image			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
FE	27.21	50.22	63.93	24.20	57.21	74.96	49.62
FE+WSSRC	**30.53**	52.65	63.05	23.63	**57.87**	**75.58**	50.55
FE+IIMC	27.21	51.54	65.70	**24.29**	57.16	74.60	50.08
FE+WSSRC+IIMC	**30.53**	54.42	64.82	23.89	57.61	74.42	50.95
FE+WSSRC+IIMC+SWSAL loss	28.54	**55.31**	**66.59**	**24.29**	57.43	75.35	**51.25**

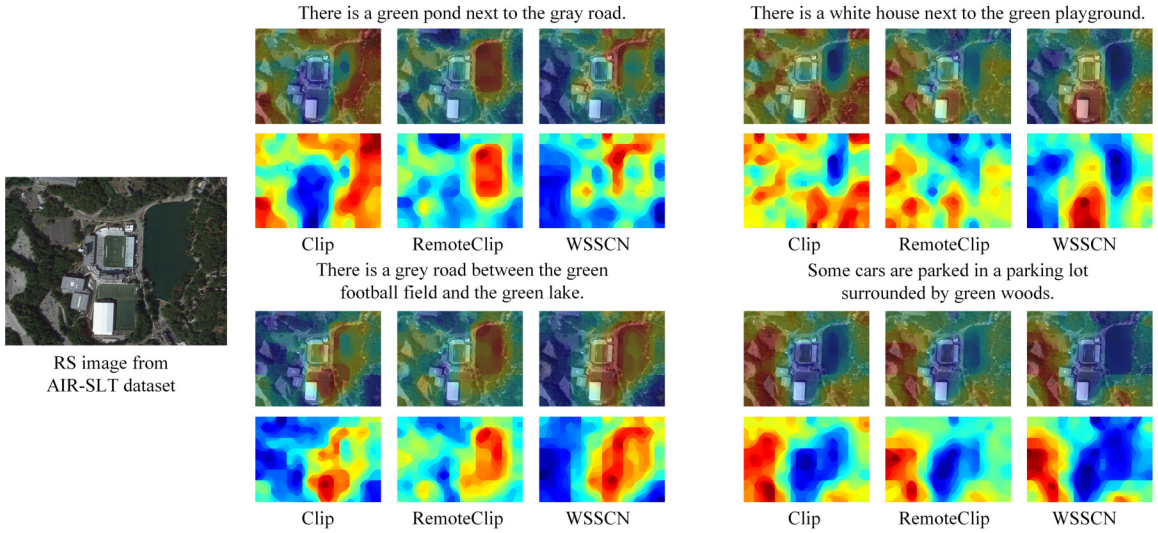


Fig. 5. Visualized qualitative comparison of SeLo on Clip, RemoteClip, and our WSSCN model.

TABLE IV
PERFORMANCE OF WSSRC

Method	RSITMD						
	Image-to-Text			Text-to-Image			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
w/o PM+DC	**29.42**	53.98	63.27	25.00	**59.07**	75.13	50.98
PM	28.76	52.43	65.27	**25.97**	58.67	**75.71**	51.14
DC	28.53	53.53	**66.59**	25.13	57.34	75.13	51.04
PM+DC	28.54	**55.31**	**66.59**	24.29	57.43	75.35	**51.25**

G. Performance of WSSRC

In this section, we perform ablation experiments to explore the preservation mechanism and the distance constraint in the WSSRC module. The experimental results are shown in Table IV, where the w/o PM + dc refers to WSSRC without applying the preservation mechanism and the distance constraint, and PM and dc represent the preservation mechanism and distance constraint, respectively. PM + dc means that the WSSRC module adopts both the preservation mechanism and the distance constraint. It can be seen from the table that the w/o PM + dc obtains the lowest retrieval accuracy since it ignores both the history information of the library and the distance constraint between different semantics. Both PM and dc improve the results by 0.31% and 0.12%, respectively. Finally, the model has made outstanding progress by adding PM and dc, because considering historical factors can build a more stable library, and limiting the distance allows the library to learn discriminable semantics.

In addition, in order to verify that the library $P = \{p_1, \ldots, p_c, \ldots, p_C\}$ contains various semantic stacks, we created several word databases $DB = \{d_1, \ldots, d_o, \ldots, d_O\}$. Each word database d_o contains words with different semantics. For instance, d_1 comprises words pertaining to colors such as yellow, purple, and orange, d_2 encompasses words with large-scale semantics like crop distribution and irrigation systems, d_6 contains words associated with relationships like near, far, and adjacent, and d_8 represents words related to quantity such as several, few, numerous, and countless. By comparing elements in P and DB, we can achieve the above verification. First, we extract one feature representation for all words in each database and then calculate the similarity matrix between the feature representations and the various semantic stacks in the library. Finally, we convert the similarity matrix into a heat map to determine which information each semantic stack focuses on, as depicted in Fig. 6. In the figure, the row vectors represent different semantic stacks $P = \{p_1, \ldots, p_c, \ldots, p_C\}$, and the column vectors represent different word databases $DB = \{d_1, \ldots, d_o, \ldots, d_O\}$. After analyzing this matrix, we discovered that each semantic stack emphasizes different types of information. For instance, since p_1 and d_6 are calculated with a higher similarity, this suggests that p_1 focuses more

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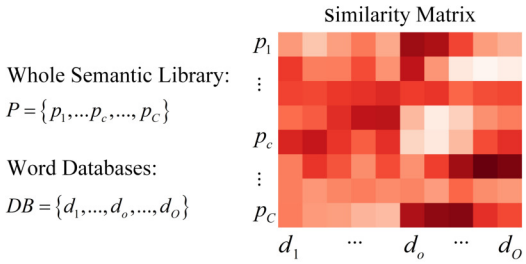


Fig. 6. Performance of whole semantic library P. The column vectors of the matrix represent different semantics, and the row vectors represent words.

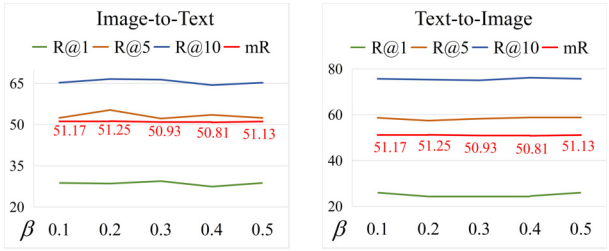


Fig. 7. Performance of hyperparameter β.

TABLE V
COMPLEXITY ANALYSIS

Type	Method	Parameters	Time(s)
N	SCAN	37M	0.0081
	CAMP	38M	0.0061
	MTFN	69M	0.0057
	SGRAF	28M	0.0085
	NAAF	23M	0.0058
R	SAM	17M	0.0474
	LW-MCR	12M	0.0024
	AMFMN	37M	0.0025
	GaLR	47M	0.0043
	MCRN	52M	0.0037
	CMFM	41M	0.0026
	HVSA	34M	0.0025
P	RemoteCLIP	427M	0.0114
	WSSCN	452M	0.0058

on the relationship between RS targets. Similarly, p_8 and d_8 exhibit higher similarity, indicating that p_8 focuses more on quantity information.

H. Complexity Analysis

We also performed complexity analysis experiments, as demonstrated in Table V. "Parameters" represents the parameters in the network, and "Time" refers to the average time needed to train the network on one sample pair. The table shows that the parameters of the large pretrained model, categorized "P," are significantly higher than those of the models not pretrained on the large model, labeled "N" and "*R*." Since our model uses the encoder in RemoteCLIP, its parameter count is slightly higher than that of RemoteCLIP. In terms of time, the proposed WSSCN has a moderate matching time of 0.0058 compared with all models. Consequently, our model strikes a balance between efficiency and effectiveness.

I. Performance of Hyperparameter β

To explore the effect of β parameters in the IIMC module, we varied β from 0.1 to 0.5 to determine the optimal values for improved model performance. The relevant results are shown in Fig. 7, and we conduct the following analysis. From the figure, we can see that, no matter the image-to-text retrieval or text-to-image retrieval, as β increases, the accuracy of the model continues to improve. When β is set to 0.2, the performance of our model reaches its peak, such as accuracy on metric mR is 51.25. After that, the precision starts to decrease, so we choose β to be 0.2 in the proposed network WSSCN.

J. Qualitative Analysis

To illustrate the effectiveness of the model more clearly, we use evaluation metrics of semantic localization (SeLo) for qualitative analysis. The SeLo is usually applied to the task of cross-modal retrieval by locating the most relevant locations in the RS image based on semantic text information. Referring to the work [61], the RS image is first cut into multiple scale slices, which are then subjected to cross-modal similarity calculation with the query. Based on regional probabilities, pixel-level averaging is applied to generate the SeLo heatmap after further noise suppression. The localized image is produced by merging the heat map into the original input image and adjusting the transparency of the heat map. In our experiment, we still follow work [61] and select the image from the AIR-SLT dataset and use its four texts as queries on Clip, RemoteClip, and our WSSCN model to calculate their SeLo results. The relevant experimental results are shown in Fig. 5, and it is evident that WSSCN is capable of detecting small objects, as demonstrated by the second query sentence in the upper right corner. This sentence contains two objects: the "white house" and the "green playground." The WSSCN model effectively locates the white house with greater accuracy. Besides, considering the third query displayed in the lower left corner, with "gray road" as the focus and involving two targets, a "football field" and a "green lake," WSSCN effectively highlights the corresponding areas. However, Clip focuses on the "playground" area and provides incorrect results, while RemoteClip pays attention to the "green lake" but overlooks the "gray road."

In addition, we also visualize the retrieval results for some of the RSITMD datasets, as shown in Figs. 8 and 9. It can be seen from Fig. 8 that for the task of image-to-text, our model can achieve correct retrieval. Although there are some wrong results, the retrieved text can also describe the image. As can be seen from (a) and (b) in Fig. 9, our model can also be well implemented for the task of text-to-image. Although (c) in Fig. 9 retrieves the wrong image, the retrieved image still fits the text description. In conclusion, from the visualization results, our model can perform the retrieval task very well.

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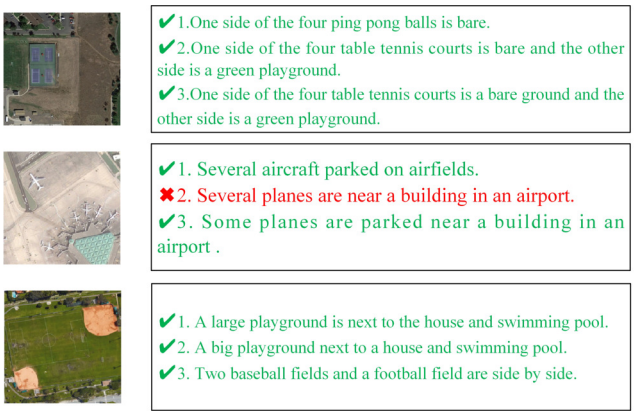


Fig. 8. Top-3 image-to-text retrieval results on RSITMD dataset. The ground-truth texts are marked with green checks, and the wrong results are indicated by cross marks.

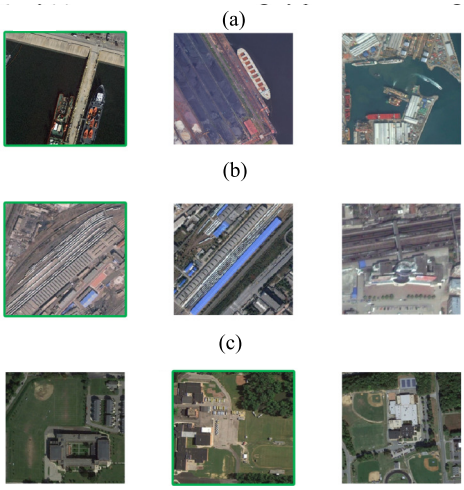


Fig. 9. Top-3 text-to-image retrieval results on the RSITMD dataset. The matched images are annotated in green boxes. (a) On both sides of the gray port were two large ships, full of red boats. (b) Black and white trains parked side by side on curved tracks. (c) School is in the forest, and there are some sports fields on the grass.

V. Conclusion

In this article, we introduce a novel WSSCN to handle the cross-modal retrieval task of RS data. A novel idea is proposed to build a complete and reliable feature description to further improve the performance of the retrieval model. Based on this, we devise three modules, including WSSRC, IIMC, and SWSAL loss, where the WSSRC module makes use of various semantic information in a whole semantic library to enable multiple semantic decoupling and a more precise and detailed feature expression, and the IIMC module to measure the intramodal and intermodal consistency from different models by reducing the bias of cross-modal saliency matrices. Finally, the SWSAL loss allows the model to adaptively focus on salient or whole semantic information based on an adjustable parameter.

We conduct numerous experiments to examine the performance of WSSCN on four large-scale RS cross-modal retrieval datasets. The experimental results demonstrate that our model outperforms the other 21 baselines. Specifically, in terms of the mR metric, WSSCN outperforms the best model, RemoteClip, by 4.93% on average across all datasets. In our future work, we will focus on two main aspects. First, we will pay attention to the interpretability of the semantic library. Since this library consists of trainable parameters, its interpretability can only be established through relevant experiments, such as Experiments 4.7 and 4.10. Second, we will concentrate on reducing the computational complexity. This involves addressing the increase in feature dimension and number of parameters after the WSSRC module, for example, by employing the mask filtering mechanism to zero out certain elements in the matrix.

References

[1] C. Bai, M. Zhang, J. Zhang, J. Zheng, and S. Chen, "LSCIDMR: Large-scale satellite cloud image database for meteorological research," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 12538–12550, Nov. 2022.

[2] K. Nogueira et al., "Exploiting ConvNet diversity for flooding identification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1446–1450, Sep. 2018.

[3] C. Bai, D. Zhao, M. Zhang, and J. Zhang, "Multimodal information fusion for weather systems and clouds identification from satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7333–7345, 2022.

[4] G. Hoxha, F. Melgani, and B. Demir, "Toward remote sensing image retrieval under a deep image captioning perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4462–4475, 2020.

[5] W. Xiong, Z. Xiong, Y. Zhang, Y. Cui, and X. Gu, "A deep cross-modality hashing network for SAR and optical remote sensing images retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5284–5296, 2020.

[6] C. Liu, J. Ma, X. Tang, X. Zhang, and L. Jiao, "Adversarial hash-code learning for remote sensing image retrieval," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 4324–4327.

[7] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," 2022, *arXiv:2204.09868*.

[8] Z. Yuan et al., "A lightweight multi-scale crossmodal text-image retrieval method in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612819.

[9] W. Kim, B. Son, and I. Kim, "ViLT: Vision-and-language transformer without convolution or region supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5583–5594.

[10] X. Tang, Y. Wang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Interacting-enhancing feature transformer for cross-modal remote sensing image and text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611715.

[11] Y. Chen, J. Huang, S. Xiong, and X. Lu, "Integrating multisubspace joint learning with multilevel guidance for cross-modal retrieval of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4702217.

[12] J. Pan, Q. Ma, and C. Bai, "Reducing semantic confusion: Scene-aware aggregation network for remote sensing cross-modal retrieval," in *Proc. ACM Int. Conf. Multimedia Retr. (ICMR)*, 2023, pp. 398–406.

[13] D. Wu et al., "Spatial–channel attention transformer with pseudo regions for remote sensing image-text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704115.

[14] Q. Cheng, Y. Zhou, P. Fu, Y. Xu, and L. Zhang, "A deep semantic alignment network for the cross-modal image-text retrieval in remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4284–4297, 2021.

[15] H. Ning, B. Zhao, and Y. Yuan, "Semantics-consistent representation learning for remote sensing image–voice retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4700614.

[16] J. Pan, Q. Ma, and C. Bai, "A prior instruction representation framework for remote sensing image-text retrieval," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 611–620.

[17] Q. Ma, J. Pan, and C. Bai, "Direction-oriented visual–semantic embedding model for remote sensing image–text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704014.

[18] Z. Ji, C. Meng, Y. Zhang, Y. Pang, and X. Li, "Knowledge-aided momentum contrastive learning for remote-sensing image text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5625213.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:12 UTC from IEEE Xplore. Restrictions apply.

[19] Z. Zhu, J. Kang, W. Diao, Y. Feng, J. Li, and J. Ni, "SIRS: Multitask joint learning for remote sensing foreground-entity image–text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5625615.
[20] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.
[21] Z. Yuan et al., "Remote sensing cross-modal text-image retrieval based on global and local information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620616.
[22] J. Zheng, M. Liang, Y. Yu, Y. Li, and Z. Xue, "Knowledge graph enhanced multimodal transformer for image-text retrieval," in *Proc. IEEE 40th Int. Conf. Data Eng. (ICDE)*, May 2024, pp. 70–82.
[23] X. Zhang, X. Li, J. An, L. Gao, B. Hou, and C. Li, "Natural language description of remote sensing images based on deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 4798–4801.
[24] X. Zhang, Q. Wang, S. Chen, and X. Li, "Multi-scale cropping mechanism for remote sensing image captioning," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 10039–10042.
[25] B. Wang, X. Lu, X. Zheng, and X. Li, "Semantic descriptions of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 8, pp. 1274–1278, Aug. 2019.
[26] Z. Shi and Z. Zou, "Can a machine generate humanlike language descriptions for a remote sensing image?," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3623–3634, Jun. 2017.
[27] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
[28] J. Li, D. Li, C. Xiong, and S. C. H. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.
[29] D. Zhu, J. Chen, X. Shen, X. Li, and M. Elhoseiny, "MiniGPT-4: Enhancing vision-language understanding with advanced large language models," 2023, *arXiv:2304.10592*.
[30] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.
[31] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "RS5M and GeoRSCLIP: A large scale vision-language dataset and a large vision-language model for remote sensing," 2023, *arXiv:2306.11300*.
[32] A.-A. Liu et al., "Text-guided knowledge transfer for remote sensing image-text retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[33] X. Zhang et al., "A fusion encoder with multi-task guidance for cross-modal text–image retrieval in remote sensing," *Remote Sens.*, vol. 15, no. 18, p. 4637, Sep. 2023.
[34] J. Nie et al., "Scale–relation joint decoupling network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412812.
[35] F. Zheng et al., "A fine-grained semantic alignment method specific to aggregate multi-scale information for cross-modal remote sensing image retrieval," *Sensors*, vol. 23, no. 20, p. 8437, Oct. 2023.
[36] S. Zhang, Y. Li, and S. Mei, "Exploring uni-modal feature learning on entities and relations for remote sensing cross-modal text-image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5626317.
[37] F. Yao et al., "Hypergraph-enhanced textual-visual matching network for cross-modal remote sensing image retrieval via dynamic hypergraph learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 688–701, 2023.
[38] K.-H. Lee, X. Chen, G. Hua, H. Hu, and X. He, "Stacked cross attention for image-text matching," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 212–228.
[39] Y. Lv, W. Xiong, X. Zhang, and Y. Cui, "Fusion-based correlation learning model for cross-modal remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6503205.
[40] Z. Zhao et al., "Masking-based cross-modal remote sensing image–text retrieval via dynamic contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5626215.
[41] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[42] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[43] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.
[44] F. Faghri, D. J. Fleet, J. Ryan Kiros, and S. Fidler, "VSE+$: Improving visual-semantic embeddings with hard negatives," 2017, *arXiv:1707.05612*.
[45] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5.
[46] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.
[47] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.
[48] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 4404119.
[49] F. Zhang, B. Du, and L. Zhang, "Saliency-guided unsupervised feature learning for scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2175–2184, Apr. 2015.
[50] Y. Huang, Q. Wu, C. Song, and L. Wang, "Learning semantic concepts and order for image and sentence matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6163–6171.
[51] Z. Wang et al., "CAMP: Cross-modal adaptive message passing for text-image retrieval," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5764–5773.
[52] T. Wang, X. Xu, Y. Yang, A. Hanjalic, H. T. Shen, and J. Song, "Matching images and text with multi-modal tensor fusion and re-ranking," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 12–20.
[53] H. Diao, Y. Zhang, L. Ma, and H. Lu, "Similarity reasoning and filtration for image-text matching," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 1218–1226.
[54] K. Zhang, Z. Mao, Q. Wang, and Y. Zhang, "Negative-aware attention framework for image-text matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15661–15670.
[55] Z. Yuan et al., "MCRN: A multi-source cross-modal retrieval network for remote sensing," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 115, Dec. 2022, Art. no. 103071.
[56] H. Yu et al., "Text-image matching for cross-modal remote sensing image retrieval via graph neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 812–824, 2023.
[57] W. Zhang et al., "Hypersphere-based remote sensing cross-modal text–image retrieval via curriculum learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5621815.
[58] G. Hu, Z. Wen, Y. Lv, J. Zhang, and Q. Wu, "Global–local information soft-alignment for cross-modal remote-sensing image–text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5623915.
[59] X. Chen, X. Zheng, and X. Lu, "Context-aware local–global semantic alignment for remote sensing image–text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5617312.
[60] R. Yang et al., "Accurate and lightweight learning for specific domain image-text retrieval," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 9719–9728.
[61] Z. Yuan et al., "Learning to evaluate performance of multi-modal semantic localization," 2022, *arXiv:2209.06515*.

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